Qier An (安琪儿)

Department of Physics and Astronomy, Johns Hopkins University
Baltimore, Maryland, USA
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Education

Ph.D. in Physics and Astronomy, Johns Hopkins University — *Aug 2024 –present*
Advisors: Prof. Timothy Brandt, Prof. Kevin Schlaufman

M.A. in Physics, University of California, Santa Barbara — *Sep 2021 –Jun 2024*
Advisor: Prof. Timothy Brandt; GPA: 3.93 / 4.00

B.S. in Engineering Physics, University of Illinois Urbana-Champaign — *Aug 2017 –Dec 2020*
Highest Honors; GPA: 3.85 / 4.00

B.S. in Mathematics and Astronomy, University of Illinois Urbana-Champaign — *Aug 2017 –Dec 2020*
Highest Honors; GPA: 3.85 / 4.00

Research

Detection and characterization of substellar objects
Astrometry, radial velocity, and direct imaging
Demographics of substellar populations
Multi-instrument data analysis and interpretation
Co-developer and maintainer of `orvara` and `htof`

Publications

Full list: ADS Library

1. **An, Q.**, Brandt, T. D., Brandt, G. M., & Venner, A. 2025, ApJS, 280, 61 — *Orbits and Masses for 156 Companions from Combined Astrometry and Radial Velocities, and a Validation of Gaia Non-single-star Solutions*
2. **An, Q.**, Lu, T., Brandt, G. M., Brandt, T. D., & Li, G. 2025, AJ, 169, 22 — *Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System*
3. Lu, T., **An, Q.**, Li, G., et al. 2025, ApJ, 979, 218 — *Planet–Planet Scattering and von Zeipel–Lidov–Kozai Migration—The Dynamical History of HAT-P-11*
4. Venner, A., **An, Q.**, et al. 2024, MNRAS, 535, 90 — *HD 28185 Revisited: An Outer Planet, Instead of a Brown Dwarf, on a Saturn-like Orbit*
5. Uyama, T., Kuzuhara, M., Beichman, C., ..., **An, Q.**, et al. 2025, AJ, 170, 272 — *Direct Imaging Explorations for Companions from the Subaru/IRD Strategic Program II; Discovery of a Brown-dwarf Companion around a Nearby Mid-M-dwarf LSPM J1446+4633*

Employment

Graduate Student Researcher, Johns Hopkins University — *Aug 2024 –present*
Graduate Student Researcher, University of California, Santa Barbara — *Sep 2021 –Aug 2024*
Teaching Assistant, University of California, Santa Barbara — *Sep 2021 –Jun 2024*
Undergraduate Researcher, University of Illinois Urbana-Champaign — *2019 –2021*

Honors, Awards, & Fellowships

AAS International Travel Grant (2025)
Dean's Graduate Fellowship, UCSB (2021)
Yee Seung Ng Scholarship, UIUC (2019)
Dean's List, UIUC (Spring 2020, Fall 2019, Spring 2018, Fall 2017)

Proposal Participation and Observing

Co-I, JWST Cycle 5 GO 10758 (PI: Yiting Li) — *accepted Mar 2026*
Co-I, NASA Keck Strategic Mission Support (PI: Thayne Currie) — *accepted Dec 2023*
Co-I, SCExAO Direct Imaging Survey, Subaru (PI: Thayne Currie) — *accepted Dec 2023*
Observer: Keck/NIRC2, Subaru/SCExAO-CHARIS (OASIS survey); reduction with pyKLIP — *2023 –present*

Conferences and Presentations

11/2026 Invited talk, Group Meeting @ CU Boulder (*Boulder, CO*)
10/2026 Invited talk, Group Meeting @ UIUC (*Urbana, IL*)
10/2026 Invited talk, Departmental Tea Talk @ Indiana University (*Bloomington, IN*)
07/2026 Talk, Emerging Researchers in Exoplanet Science Symposium XI (*Ann Arbor, MI*)
07/2026 Talk, HotSci @ JHU/STScI (*Baltimore, MD*)
04/2026 Talk, The Brown Dwarf to Exoplanet Connection (*Newark, DE*)
03/2026 Talk, Preparing for Gaia's Exoplanet Bonanza Workshop (*Bavaria, Germany*)
12/2025 Invited talk, Departmental Journal Club @ TDLI (*Shanghai, China*)
12/2025 Poster, International Conference on Exoplanets and Planet Formation (*Shanghai, China*)
09/2025 Talk, Departmental Astro Coffee @ JHU (*Baltimore, MD*)
09/2025 Poster, G-WiP @ JHU Annual Summit (*Baltimore, MD*)
07/2025 Poster, Sagan Exoplanet Workshop (*Pasadena, CA*)
06/2025 Talk, Emerging Researchers in Exoplanet Science Symposium X (*Princeton, NJ*)
07/2024 Poster, Sagan Exoplanet Workshop; travel grant awarded (declined for virtual) (*Pasadena, CA*)
07/2023 Poster, Sagan Exoplanet Workshop; hands-on session helper; travel grant (*Pasadena, CA*)
07/2022 Poster, Sagan Exoplanet Workshop; hands-on session helper; travel grant (*Pasadena, CA*)
05/2022 Attendee, Exoplanets IV (*Las Vegas, NV*)

Teaching

University of California, Santa Barbara

PHYS 3L: Calculus-based Physics Lab (2021), Teaching Assistant

PHYS 29: Intro to Computer Programming (2022–2023), Teaching Assistant

ASTRO 1: Basic Astronomy (2023), Teaching Assistant

PHYS 129L: Intro to Scientific Computation (2024), Teaching Assistant

Community Service

2026 – Referee, Monthly Notices of the Royal Astronomical Society (MNRAS)
11/2025 SOC, STScI/JHU ExoJamboree
09/2025 LOC, G-WiP @ JHU 2025 Annual Summit
2025 – Referee, American Astronomical Society (AAS) journals
2022, 2023 Hands-on session helper, Sagan Exoplanet Workshop

Skills

Python (NumPy, SciPy, astropy, emcee/ptemcee, pyKLIP), orbit fitting (`orvara`, `htof` co-developer and maintainer), Gaia/Hipparcos astrometry, radial velocities, high-contrast imaging (SCExAO/CHARIS, NIRC2), HST/STIS coronagraphy, hierarchical Bayesian inference, LaTeX. Languages: English, Mandarin Chinese.

CV last updated: October 27, 2026